

(1.) Bath normalizacija in razlagi

$$D(\epsilon) = \frac{1}{\epsilon - \omega + i\gamma} - \frac{1}{\epsilon + \omega + i\gamma} = \frac{2\omega}{(\epsilon + i\gamma)^2 - \omega^2} = \frac{2\omega(\epsilon^2 - (\omega^2 + \gamma^2) - 2i\epsilon\gamma)}{(\epsilon^2 - (\omega^2 + \gamma^2) + 2i\epsilon\gamma)(\epsilon^2 - (\omega^2 + \gamma^2) - 2i\epsilon\gamma)}$$

$$= \frac{2\omega(\epsilon^2 - (\omega^2 + \gamma^2) - 4i\epsilon\gamma)}{(\epsilon^2 - (\omega^2 + \gamma^2))^2 + 4\epsilon^2\gamma^2} \rightarrow A(\epsilon) \propto \text{Im} D = \frac{-4\epsilon\omega\gamma}{(\epsilon^2 - (\omega^2 + \gamma^2))^2 + 4\epsilon^2\gamma^2}$$

podobi strani:

$$-\frac{1}{\pi} \text{Im}(D) = -\frac{1}{\pi} \left(\frac{-\gamma}{(\epsilon - \omega)^2 + \gamma^2} - \frac{-\gamma}{(\epsilon + \omega)^2 + \gamma^2} \right) = \mathcal{L}_\gamma(\epsilon - \omega) - \mathcal{L}_\gamma(\epsilon + \omega)$$

$$\Leftrightarrow A(\epsilon) \propto \mathcal{L}_\gamma(\epsilon - \omega) - \mathcal{L}_\gamma(\epsilon + \omega) \quad \int d\epsilon A(\epsilon, \omega) = 1 \text{ (normalizacija)}$$

$$A_\pm(\epsilon) \equiv \frac{\mathcal{L}_{\omega_\pm}(\epsilon - \omega_\pm) - \mathcal{L}_{\omega_\pm}(\epsilon + \omega_\pm)}{\frac{2}{\pi} \arctan(\frac{\omega_\pm}{\epsilon_\omega})}, \quad \mathcal{L}_{\omega_\pm}(x) = \frac{\epsilon_\omega}{\pi(x^2 + \epsilon_\omega^2)}, \quad A(-\epsilon) = -A(\epsilon)$$

$$1 + N(-\epsilon) = -N(\epsilon)$$

① $\epsilon > 0$: $A(\epsilon)(1 + N(\epsilon)) \checkmark$ (anis.) $\nearrow$ $\epsilon = -\omega$

② $\epsilon < 0$: $A(-\epsilon)(1 + N(-\epsilon)) = A(\epsilon)N(\epsilon)$, (abs.) $\nearrow$
 $\hat{\epsilon} = -\epsilon > 0$

$$\rightarrow A_{\epsilon \rightarrow 0} = \delta(\epsilon - \omega_0) - \delta(\epsilon + \omega_0)$$

drugače: $A(|\epsilon|) \rightarrow \delta(|\epsilon| - \omega_0) - \delta(|\epsilon| + \omega_0) \xrightarrow{\omega_0 \rightarrow 0} \delta(|\epsilon| - \omega_0)$ $\delta(|\epsilon| - \omega_0) = \delta(\epsilon + \omega_0) + \delta(\epsilon - \omega_0)$

$$\rightarrow A(|\epsilon|)(\delta(-\epsilon) + N(|\epsilon|)) \rightarrow \delta(|\epsilon| - \omega_0)(\delta(-\epsilon) + N(|\epsilon|))$$

$\rightarrow$ ① $\epsilon > 0$: $\delta(\epsilon - \omega_0)N(\epsilon)$

② $\epsilon < 0$: $\delta(\epsilon + \omega_0)(1 + N(\epsilon))$

ok.

$\Rightarrow$ Bath rate je kontribucija k:

$$\Gamma_{\mu k \rightarrow \nu l}^{(T)} = \Gamma_{\mu k}^{\mu\nu} = 2\pi |G_{\mu\nu}(k_2)|^2 \cdot A(|E_{\mu k} - E_{\nu l}|, \omega_2) \cdot (G(E_{\mu k} - E_{\nu l}) + N(|E_{\mu k} - E_{\nu l}|, T))$$

$p = k \rightarrow l$

(in) $|G_{\mu\nu}|^2 = \left| \mathcal{U}_{k+\frac{z}{2}}^T \begin{pmatrix} g_{aa} & g_{ab} \\ g_{ba} & g_{bb} \end{pmatrix} \mathcal{U}_k \right|^2$; $g_{aa}, g_{bb} = \text{intra band}$
 $g_{ab}, g_{ba} = \text{inter band}$

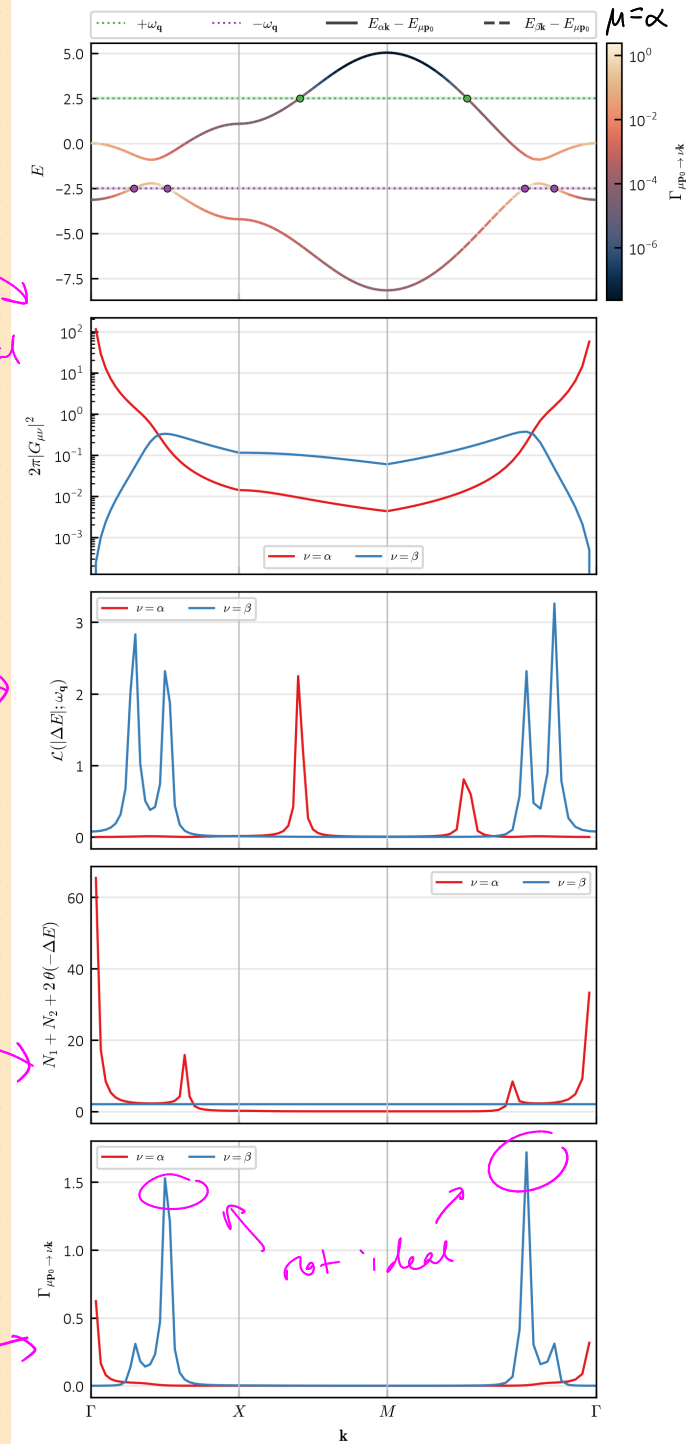
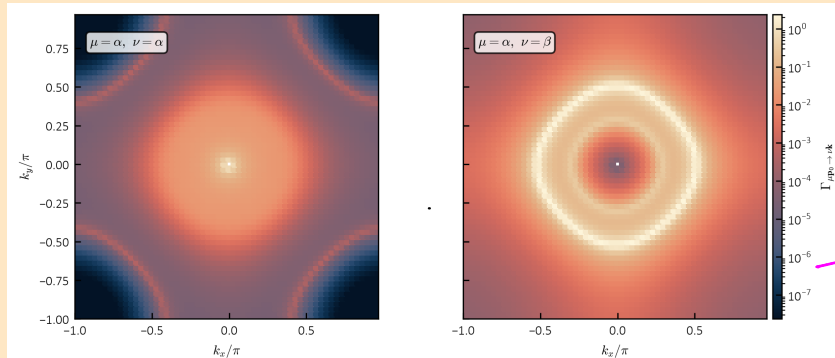
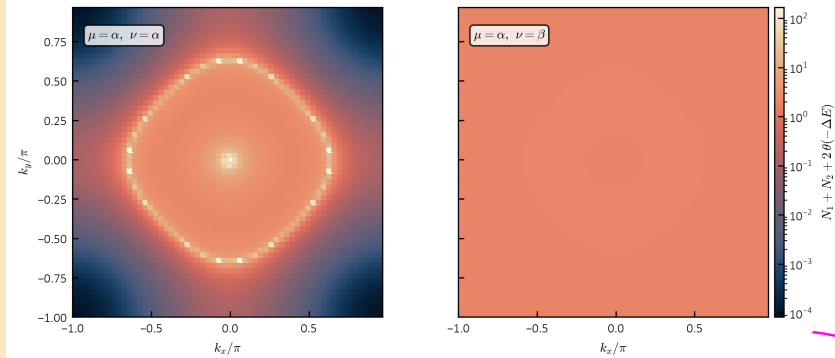
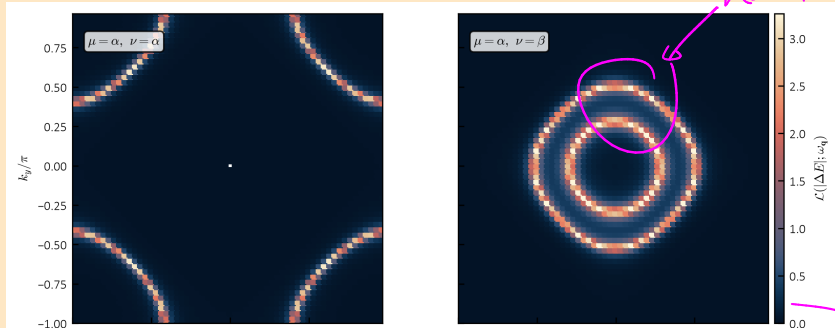
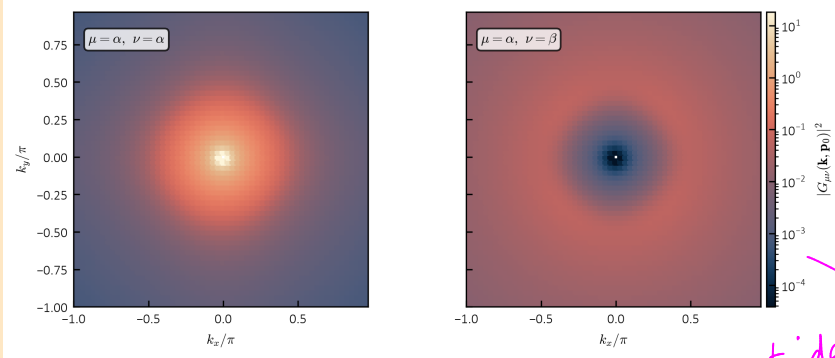
$g_{ab} = \frac{-iC_{ab}}{\sqrt{\omega_2}} \cdot \left(\frac{2}{2^L + \left(\frac{1}{\lambda}\right)^2} \right)$ $\vec{z} \cdot \vec{z}^* \approx 2$
 $\frac{1}{\lambda} = 0.001$

→ ④ intra band only: $g_{\alpha\alpha} = g_{\alpha\beta} = 0$.

① $\omega_{\vec{g}} = \omega_0 = \text{optical branch}$, fix: $\vec{p}_0 = (0, 0)$

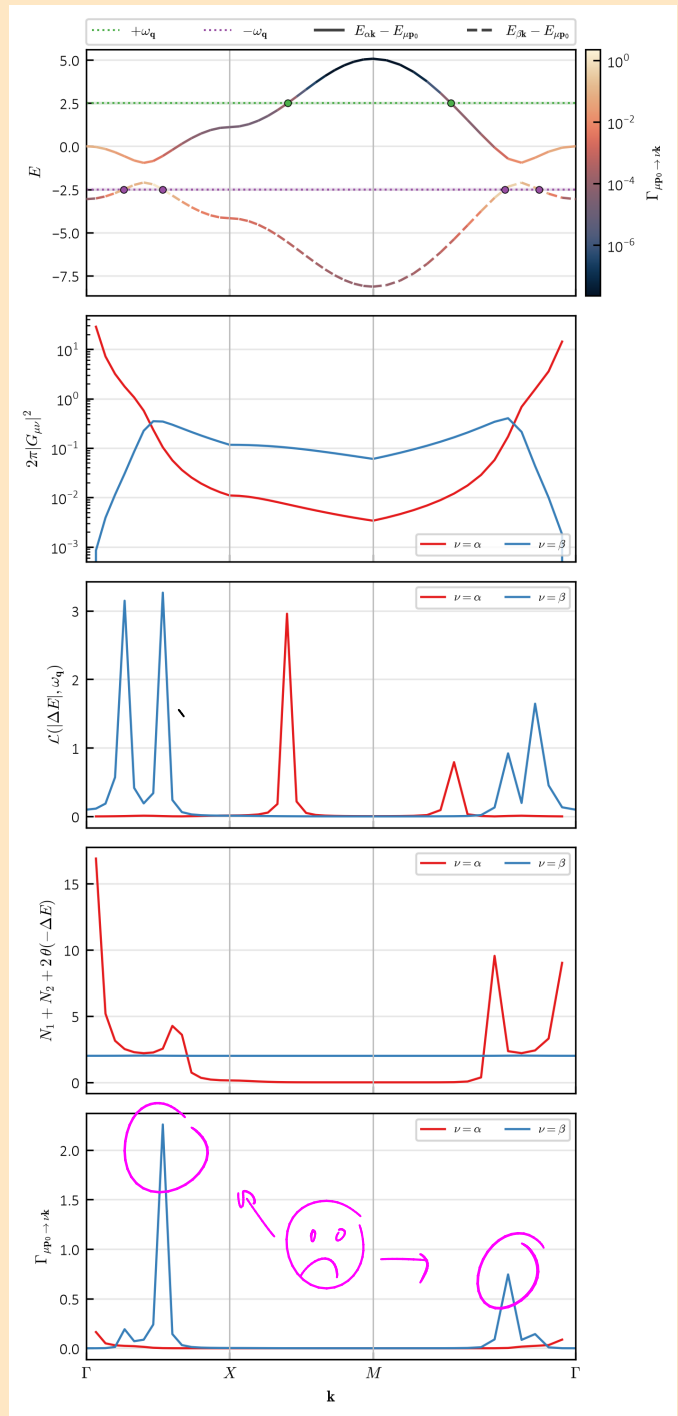
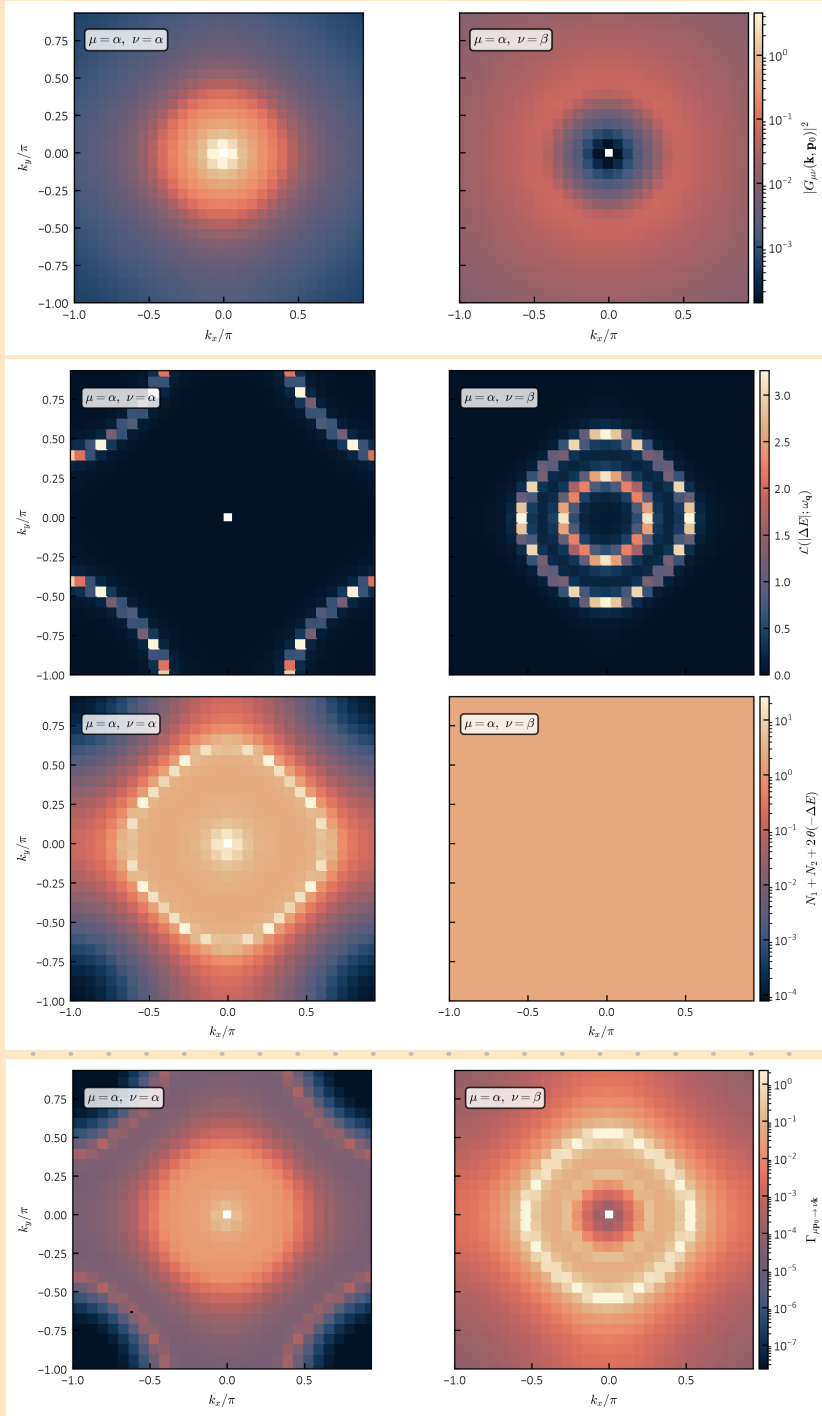
↳ $\mu = \alpha$: $\Gamma_{\alpha p_0 \rightarrow \nu k}$:

$N_k = 60 \times 60$ grid, $\frac{1}{2} = 0.001$, $\gamma = 0.1$, $\omega_0 = 2.5$



d=0.57638, err=7.93e+07, m=0.72011

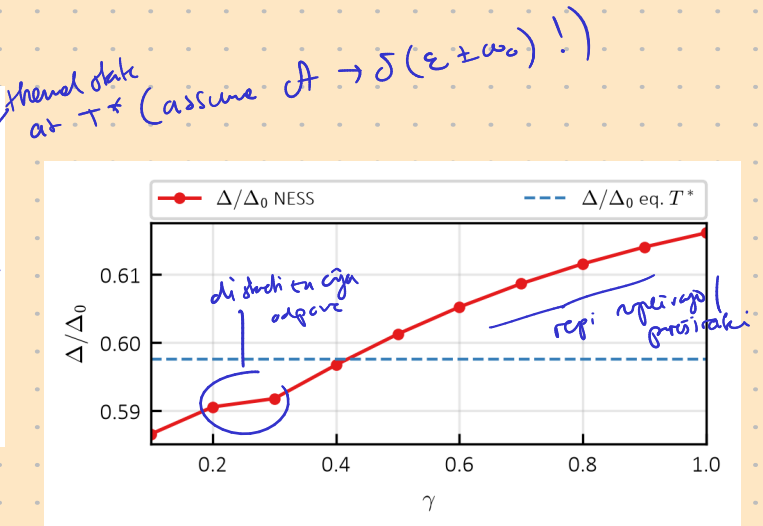
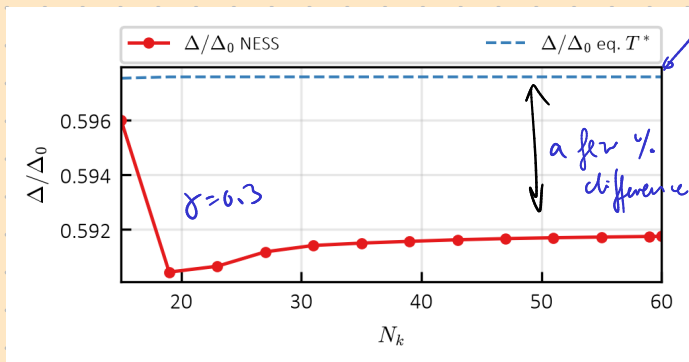
→ lower resolution: (b)



$\Delta = 0.57002$, $\text{err} = 7.57e-06$, $m = 0.71982$
 Δ "na-ho"

← this does affect steady state, how much?

→ discretization effects on NESS:

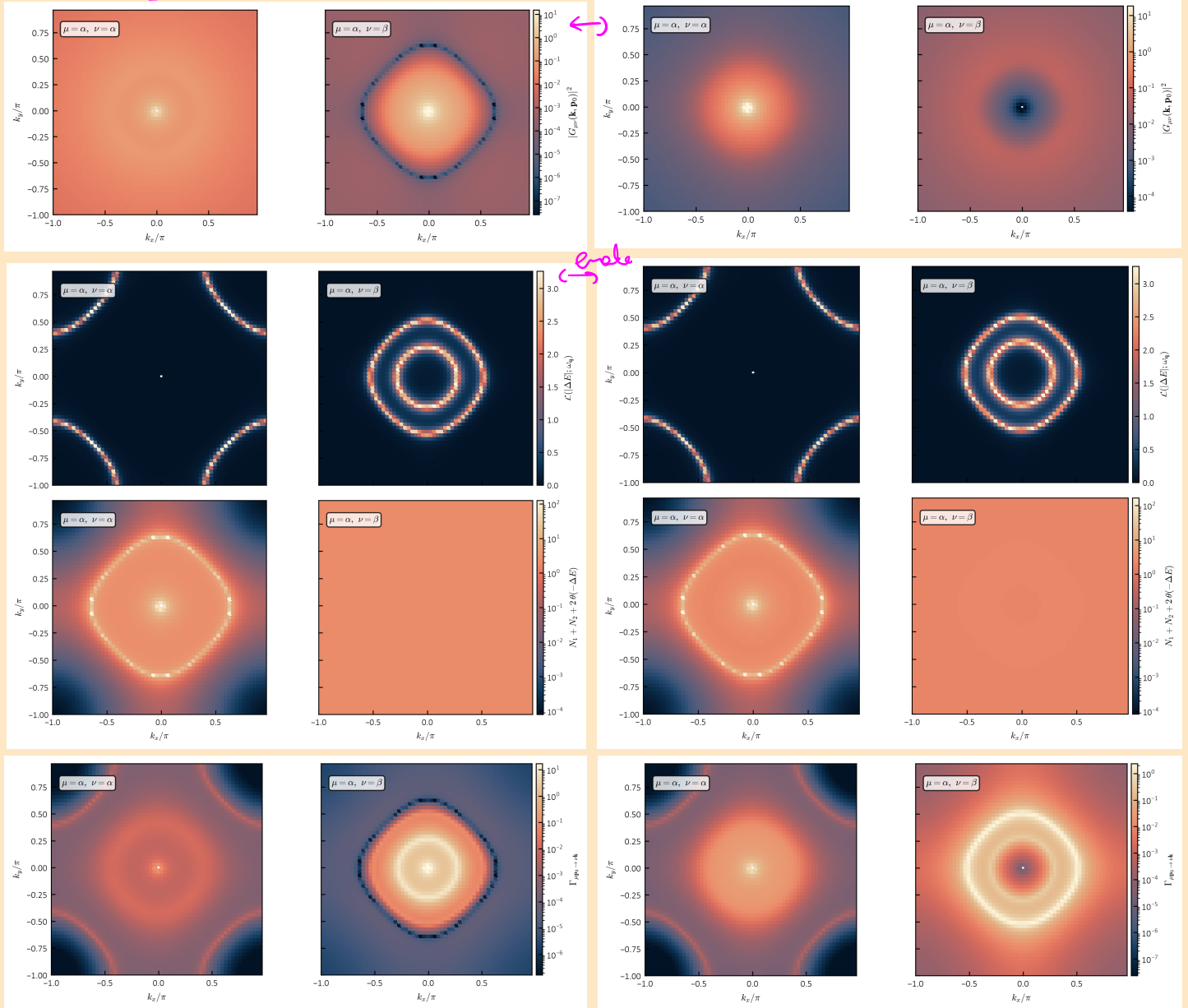


② interband only: $c_{ab} = c_{ba} = 1$, $c_{aa} = c_{bb} = 0$

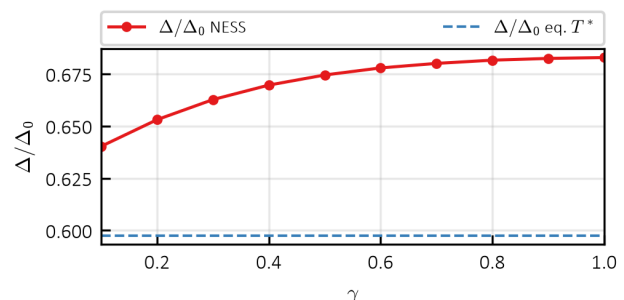
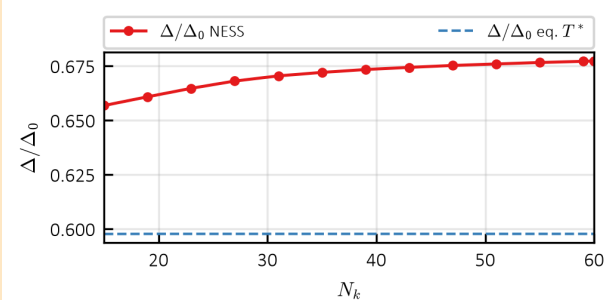
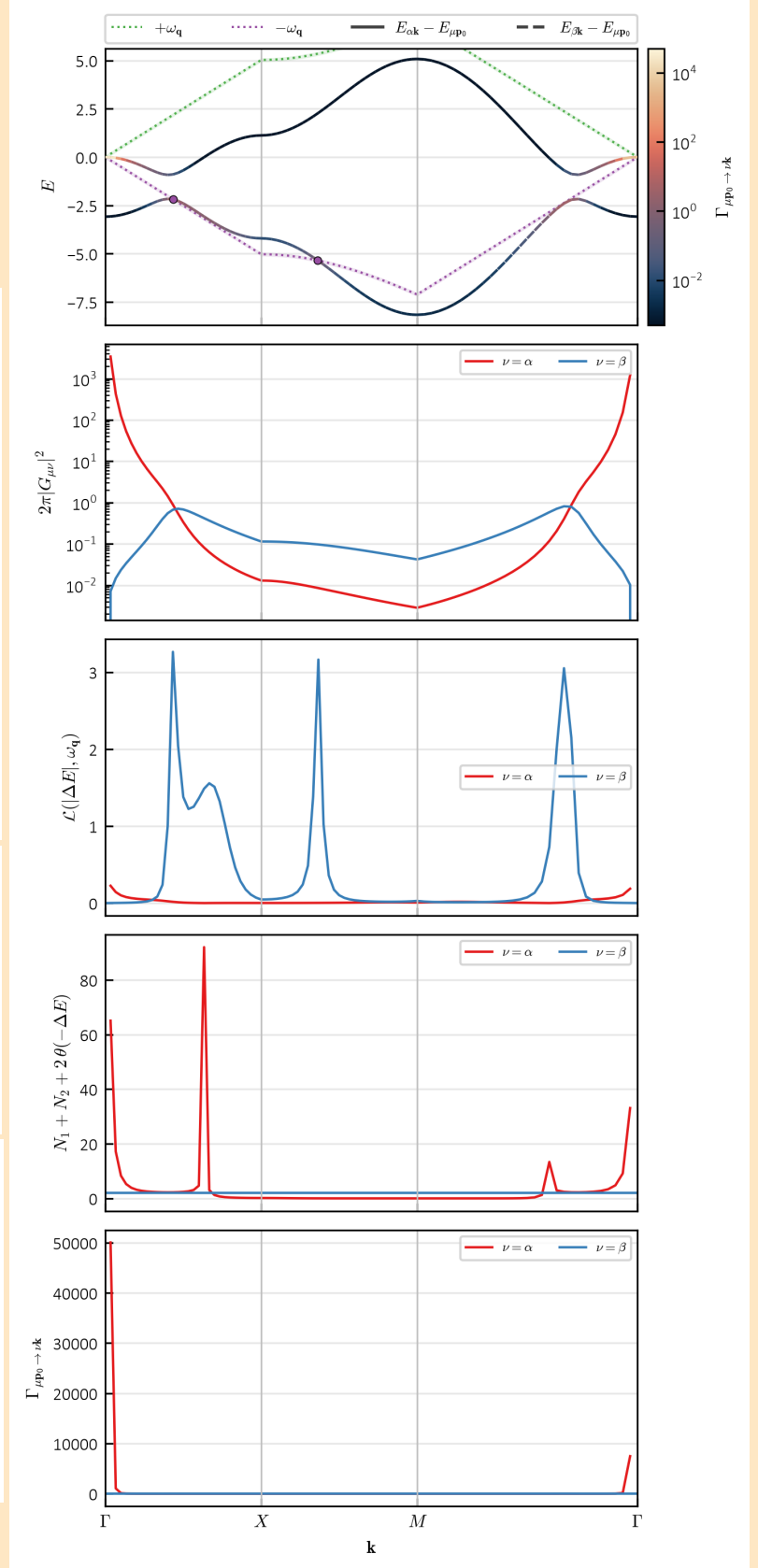
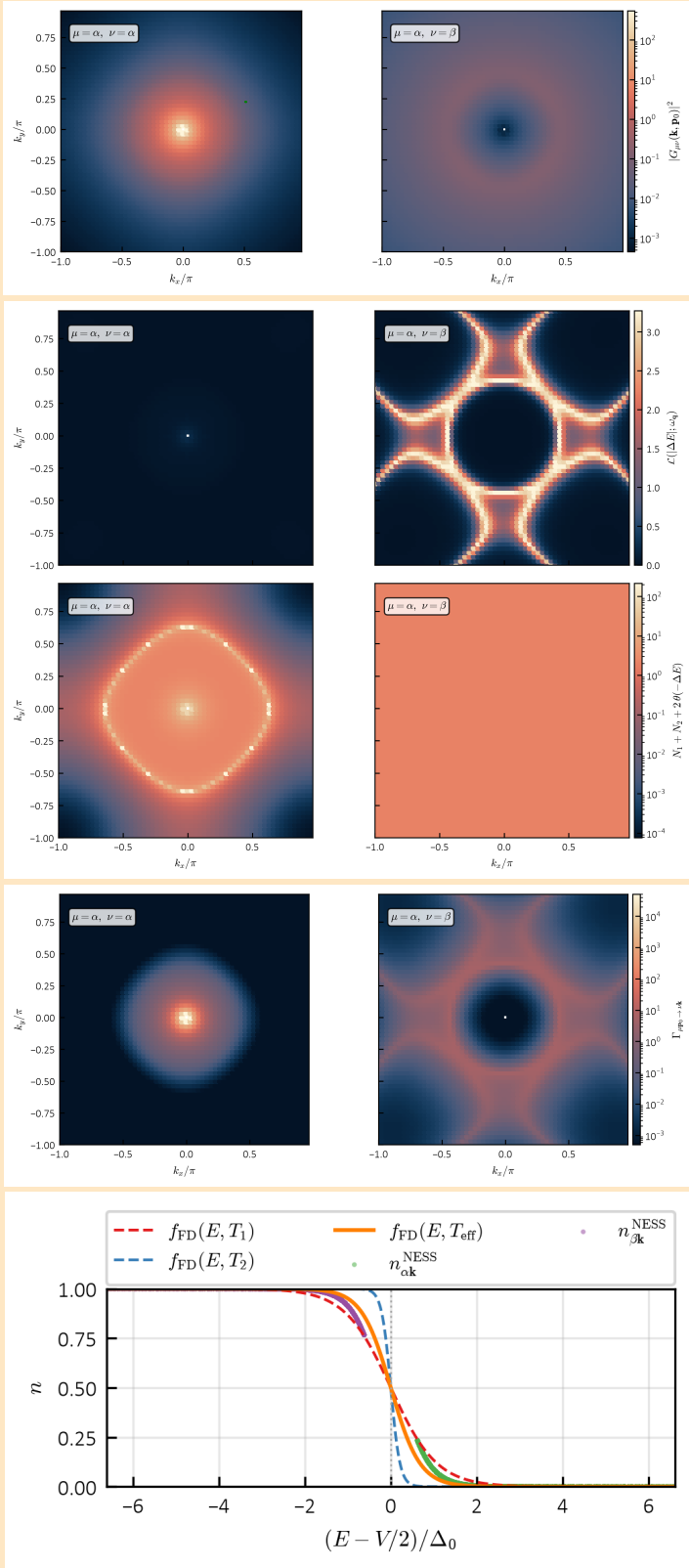
③ → enaki parametri kot pri ①, razen $c_{ab} \dots$ → spremenim se samo $|c_{\mu\nu}(k, p)|^2$

③ INTERBAND

④ INTRA.

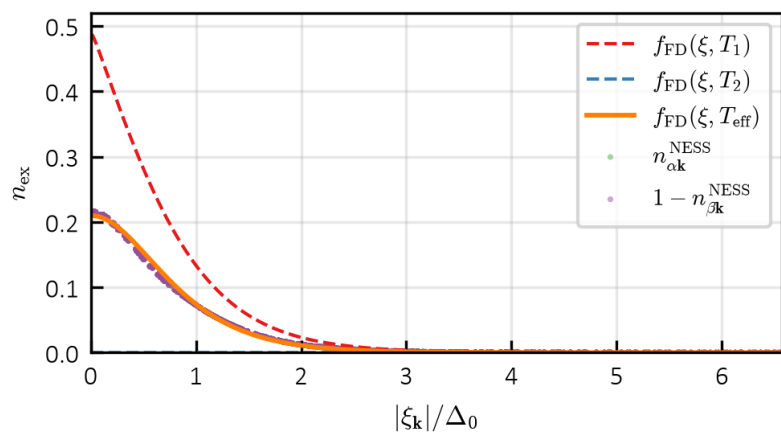
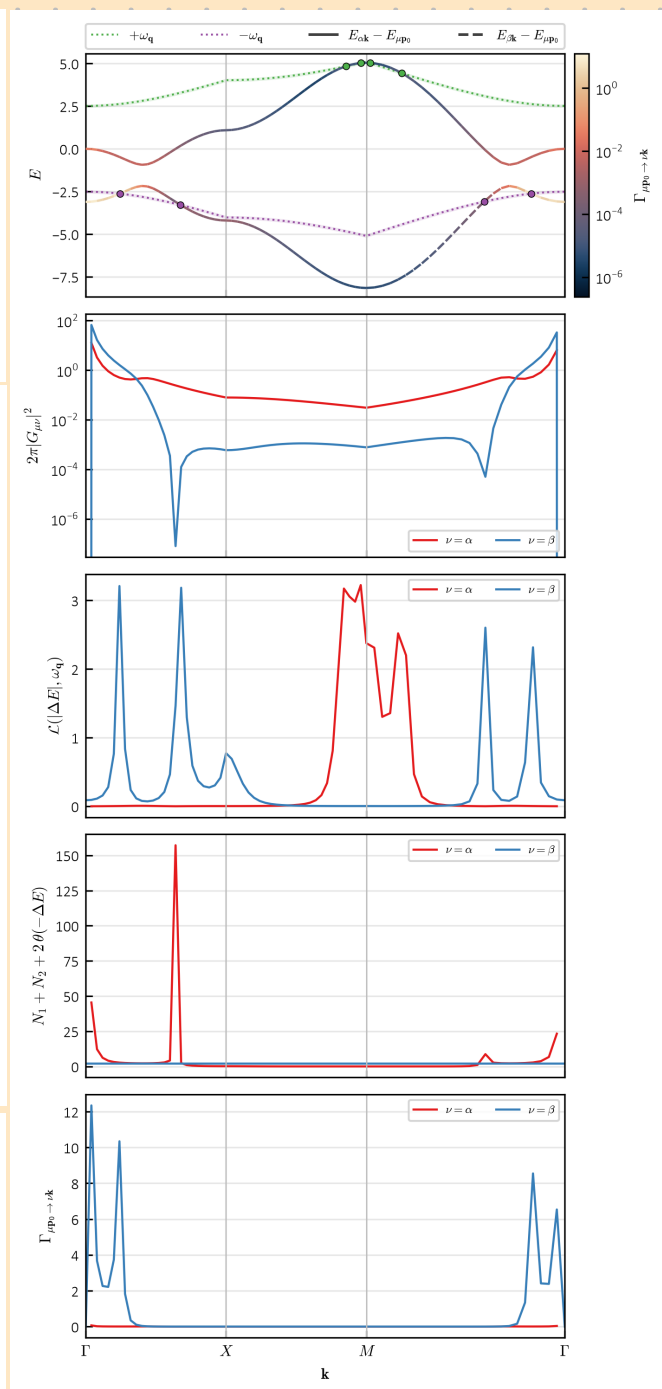
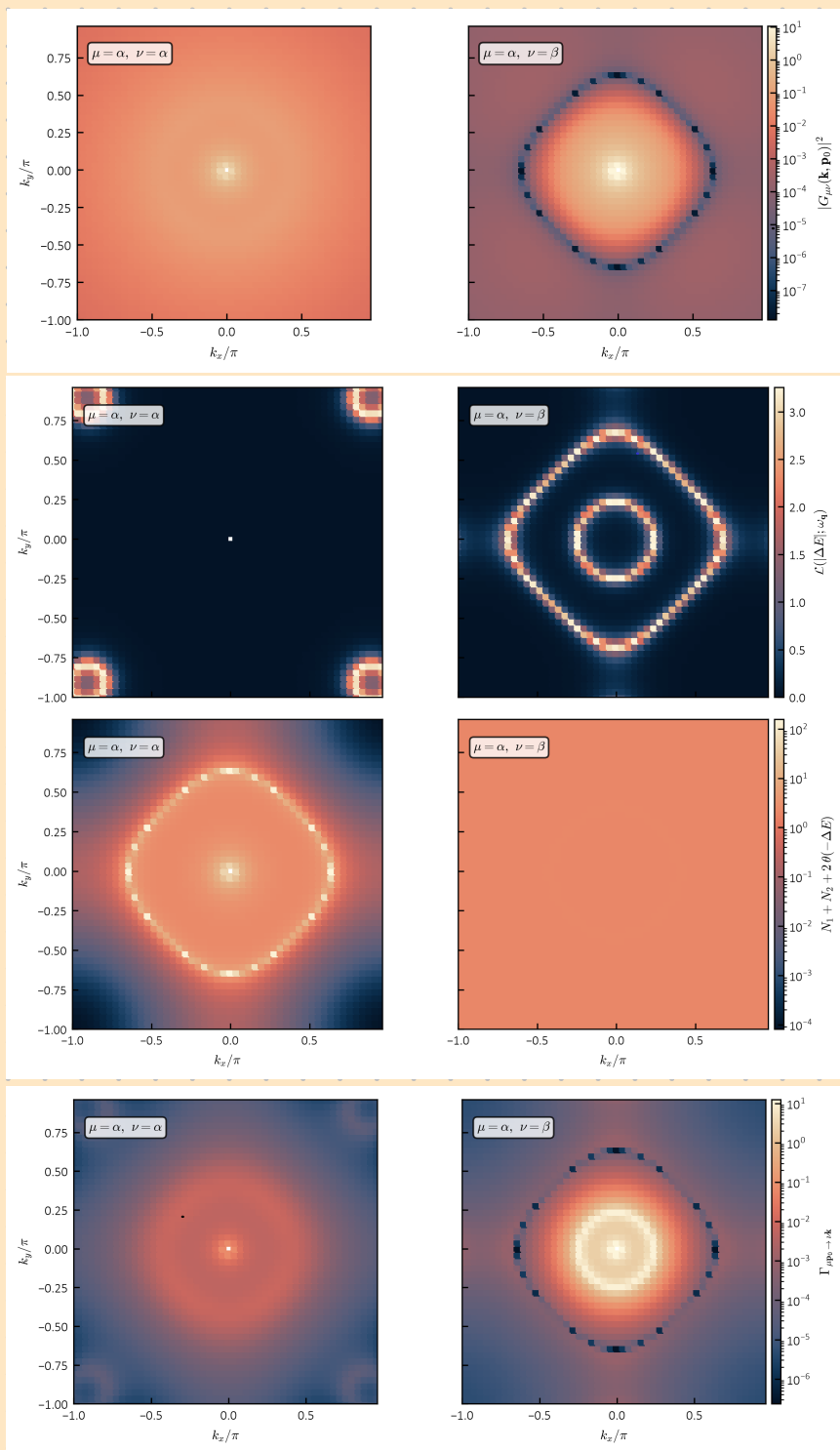


3. A kvantni fenomeni, $\omega_z = c|\mathbf{k}|$, $c = 1.6$, integrirani only. $\rightarrow$ stanja ni so rečevali: $T_2^*(T_1, T)$!

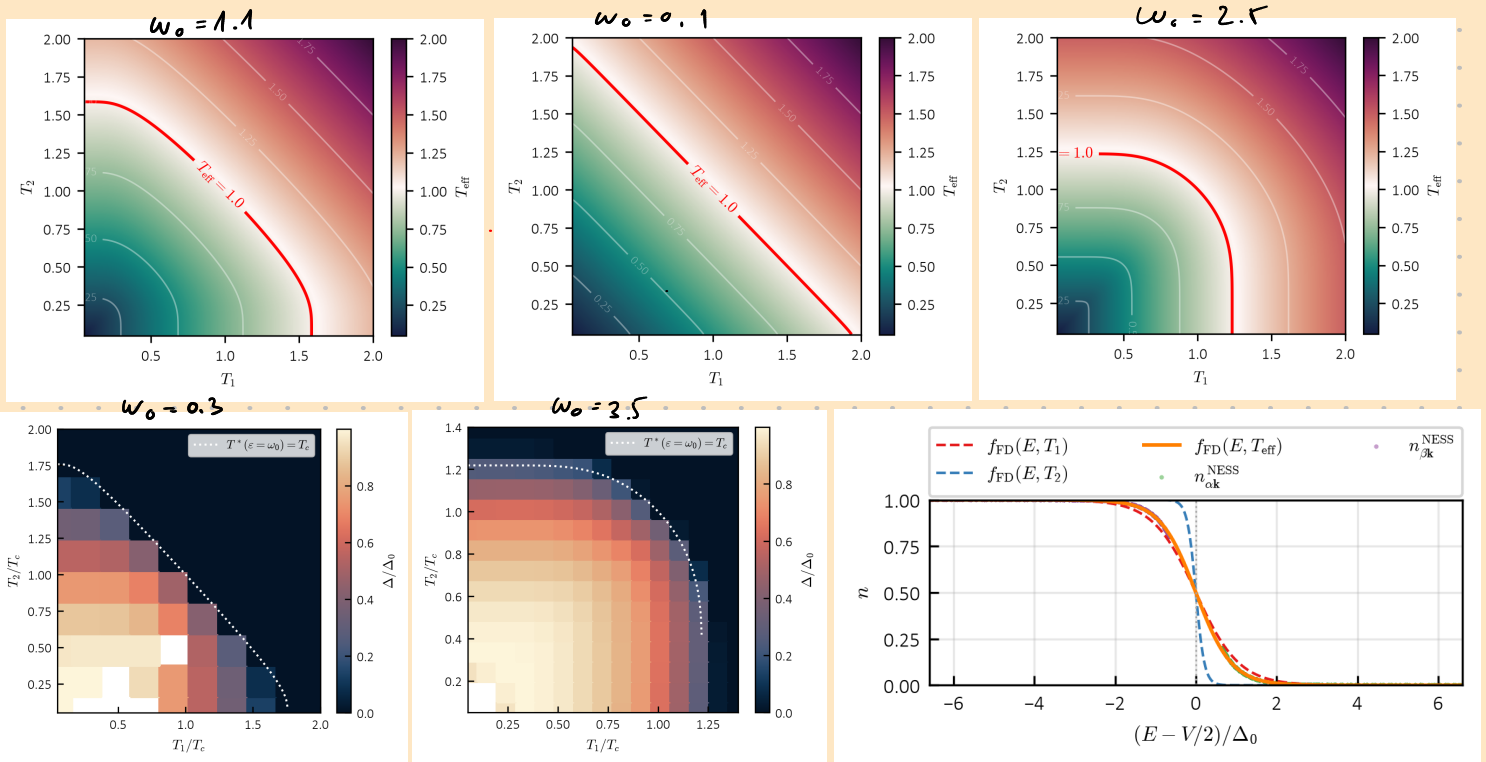


4) Resonator primer ($\underline{c} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$, interband)

→ paralelna so met $\sim$ thermal, leži na T^* njene $T_{\text{eff}} =$ temperatura obliki kopeli skupaj.



② Equal baths in analytical limits za $\Delta(T_1, T_2)$, $\omega_0 = \text{const.} = \omega_0$.



→ stanja so ~ thermal z neko temperaturo $T_{\text{eff}} = T^*$: $\frac{\Gamma(\epsilon)}{\Gamma(-\epsilon)} = e^{\epsilon/T^*(\epsilon)}$

→ za dve enaki kopeli:

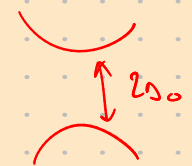
$$\frac{\Gamma(\epsilon)}{\Gamma(-\epsilon)} = \frac{2 \left(1 + \frac{N_1 + N_2}{\epsilon} \right)}{2 \frac{N_1 + N_2}{\epsilon}} = \frac{1 + \bar{N}}{\bar{N}} = e^{\epsilon/T^*}$$

$$\Rightarrow T^*(\epsilon) = \frac{\epsilon}{\ln \left(1 + \frac{1}{\bar{N}} \right)} \quad \rightarrow \text{assumes da je resonanca okoli } \epsilon \approx \omega_0$$

$$\rightarrow T^*(\omega_0) = \frac{\omega_0}{\ln \left(1 + \frac{1}{\bar{N}(\omega_0, T_1, T_2)} \right)}$$

$\uparrow$ $T^* = T_c$

$$\rightarrow T_2(T_1, \omega_0) \approx \frac{\omega_0}{\ln \left(1 + \left[\frac{2}{e^{\omega_0/T_c} - 1} - \frac{1}{e^{\omega_0/T_1} - 1} \right]^{-1} \right)}$$

V PRÁPADNĚ: inicializace při $T=0 \rightarrow \Delta_0$ 

ěe imos točno $\delta(\epsilon \pm \omega_0) \rightarrow$ ěc $\omega_0 < 2\Delta_0$ potom izolator
hopeli ne ěhi. $\Leftrightarrow \Delta = \Delta_0 > 0 \forall (T_1, T_N)$, ěc sgrevame!

$\rightarrow$ a j to fěiěkalno?

$\rightarrow$ ěc p $\delta \rightarrow A$ (mudi nabo repov) $\rightarrow$ pěsai relakivna.

3. Nesimetrične kopeli: notinja $\varepsilon = E_{\mu, L} - E_{\nu, R}$, $\rho_{\nu, \mu} = 0$, konstantno disperzija ω .

ena kopel: $\Gamma_{\mu\nu}^{(\mu)} = \Gamma_{\nu\mu}^{(T)} = 2\pi |G_{\mu\nu}(\vec{k}_{\mu, R})|^2 \cdot A(|\varepsilon|, \omega) [\theta(-\varepsilon) + N(|\varepsilon|, T)]$

$\varepsilon > 0: A(|\varepsilon|, \omega) N(\varepsilon, T)$
 $\varepsilon < 0: A(|\varepsilon|, \omega) [1 + N(\varepsilon, T)] \sim \delta(\pm \varepsilon - \omega) \checkmark$

Za dve različni kopeli, imamo:

$$\Gamma_{\mu\nu}^{(\mu)} = \Gamma_{\mu\nu}^{(T_1)} + \Gamma_{\mu\nu}^{(T_2)} = 2\pi \left\{ |G_{\mu\nu}|^2 \left[A(|\varepsilon|, \omega_1) (\theta(-\varepsilon) + N(|\varepsilon|, T_1)) + A(|\varepsilon|, \omega_2) (\theta(-\varepsilon) + N(|\varepsilon|, T_2)) \right] \right\}$$

$$e^{\varepsilon/T^*(\varepsilon)} = \frac{\Gamma(\varepsilon)}{\Gamma(-\varepsilon)} = \frac{\Gamma_{\downarrow} - \text{emisija}}{\Gamma_{\uparrow} - \text{absorpcija}} = \frac{a_1(\varepsilon)(1+N_1) + a_2(\varepsilon)(1+N_2)}{a_1(\varepsilon)N_1 + a_2(\varepsilon)N_2}$$

$$a_i = \frac{A_i(\varepsilon)}{\omega_i}$$

če v neki točki (T_1, T_2) bo $\Delta > 0$, odloča T^* temperatura, ki jo čuli transfer pri ε .

$$T^*(\varepsilon) = \frac{\varepsilon}{\ln\left(1 + \frac{1}{N}\right)}$$

$$\bar{N} = \frac{a_1}{a_2 + a_1} N_1 + \frac{a_2}{a_1 + a_2} N_2$$

$T^* = T_c = 1$ (pogoji)
 $\bar{N} = N(\varepsilon, T_c) = p N_1 + (1-p) N_2 \rightarrow N_2 = \frac{\bar{N} - p N_1}{1-p} = \frac{1}{e^{\varepsilon/T_c} - 1}$

$$\Rightarrow T_2(T_1, \varepsilon) = \frac{\varepsilon}{\ln\left(1 + \left(\frac{N(\varepsilon, T_c) - p N(\varepsilon, T_1)}{1-p}\right)^{-1}\right)}$$

→ Poskus fiksirati ε^* , kjer je prehod:

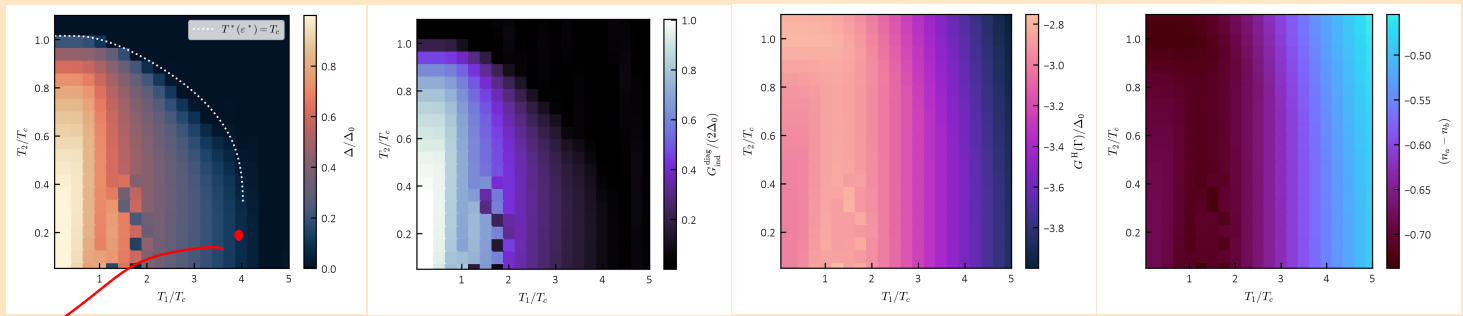
obziroma robni gaza imamo tranzicije, z energijo ε , ki čuti na efektivno temperaturo: T^* (glej plot). → $\exists$ neka drena, kjer je $T^* \leq T_c$, ε^* je max. energija, ki zadostuje temu drena.

→ Tranzicijska drena = $T_2(T_1, \varepsilon^*)$

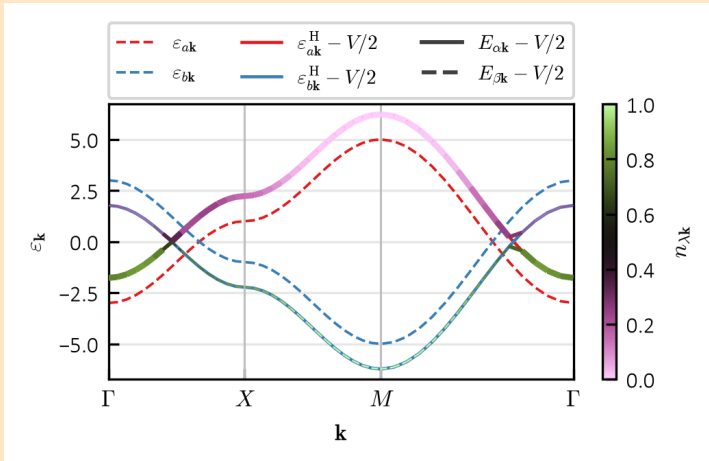
podla dolžino reza

Primer: $w_1 = 3.5$
 $w_2 = 1.0$

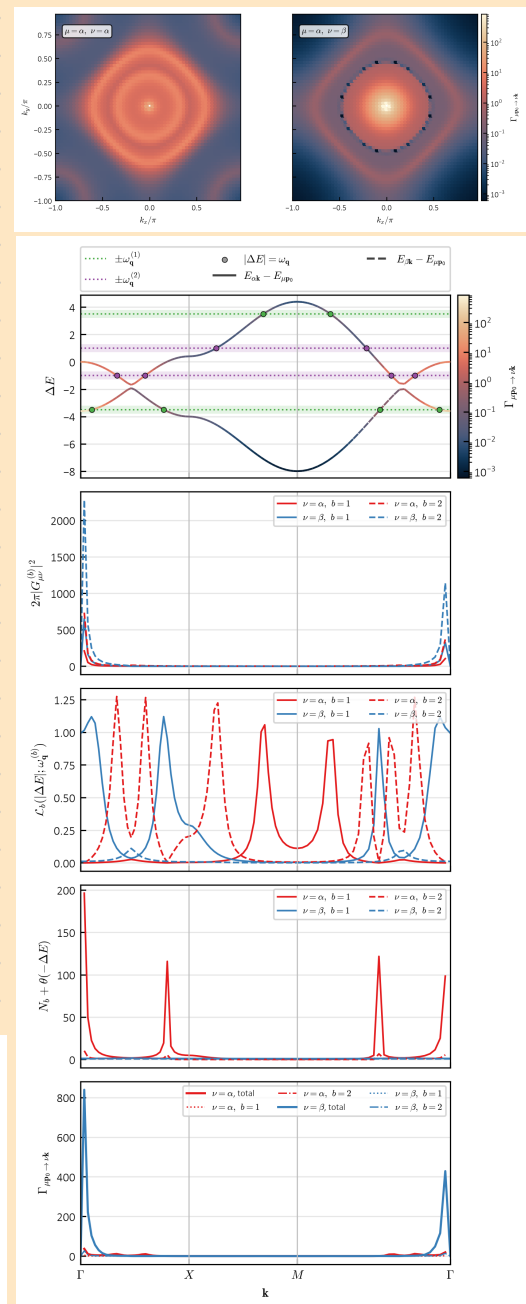
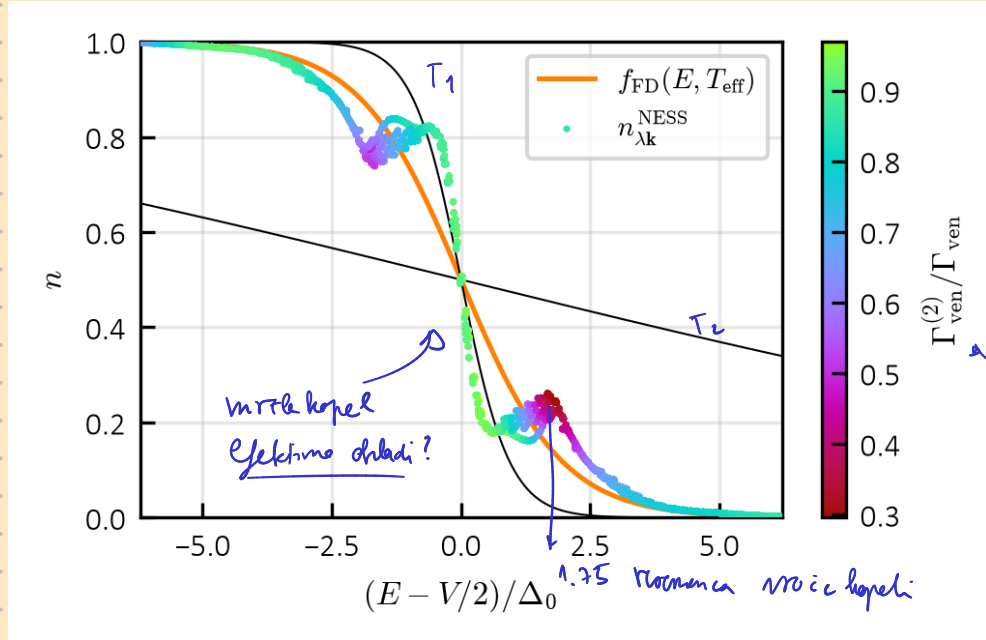
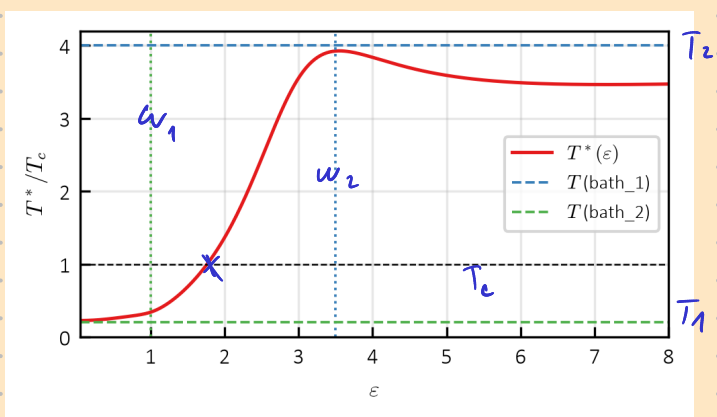
$\Gamma_{out, \beta_0}^{(b)} = \sum_k \Gamma_{\beta \rightarrow k}^{(b)}$, perimetro: $\underline{C} = \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}$



$T_1 = 4, T_2 = 0.2$ (bath 2 = cold, bath 1 = hot)

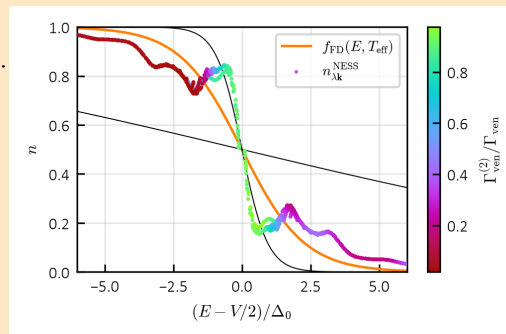


T^* , ki jo vidi energy transfer ε :

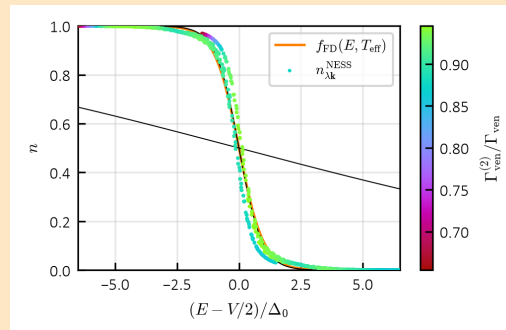


delež mitnih / (mrež + toplin)
 prehodov iz stanja
 $(\sum_p \Gamma_p^{(b)})$
 $2E = w_2 \Rightarrow E = 1.75$

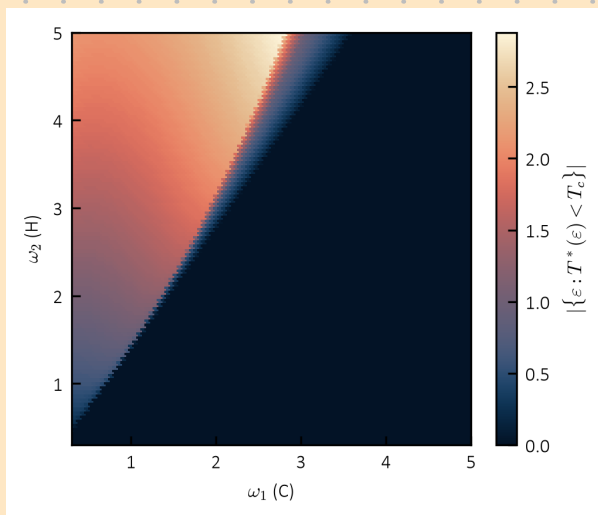
$$\rightarrow \hat{c}_c \hat{c} = \begin{pmatrix} 0 & 2 \\ 2 & 0 \end{pmatrix}$$



$$\rightarrow \hat{c} \hat{c} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$



$\rightarrow g_{\mu\nu} \sim 0$ rather than $\hat{z} \sim \hat{\sigma}$
 $\hookrightarrow g_{xx} = g \cdot (\mu_k \mu_p + v_p v_k) =$
 $= g \cos\left(\frac{\theta_k - \theta_p}{2}\right) \sim g$
 $g_{xy} = g (\mu_k v_p - \mu_p v_k) =$
 $= g \sin\left(\frac{\theta_k - \theta_p}{2}\right) \sim 0$

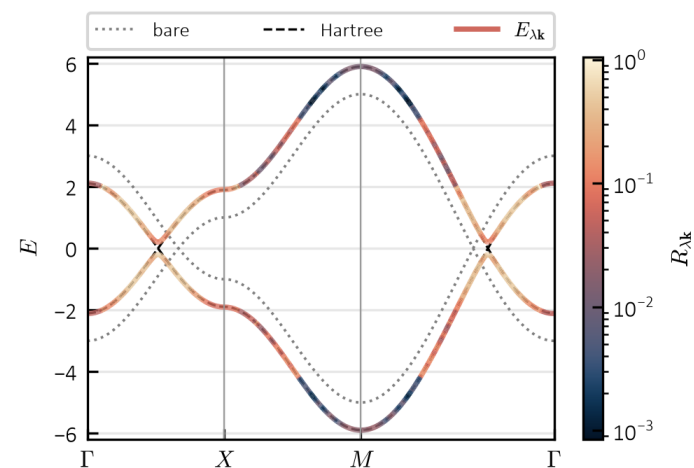
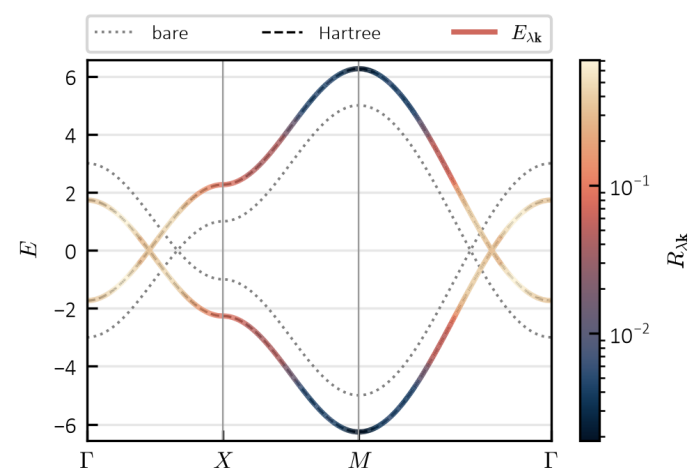
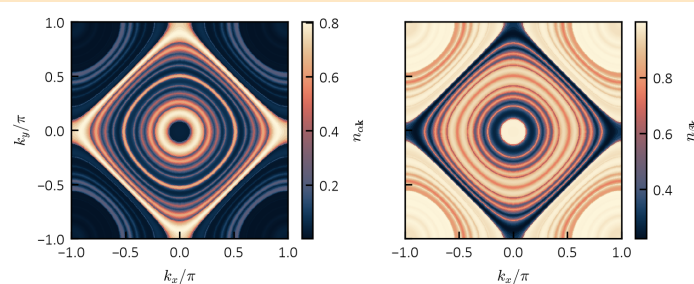
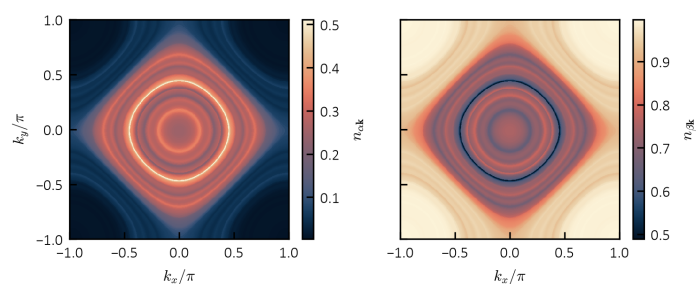
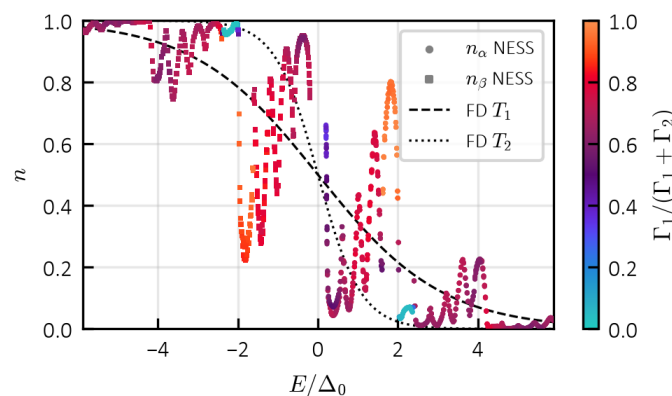
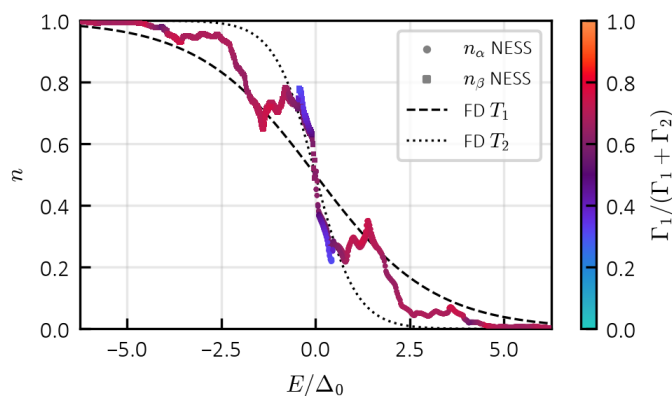
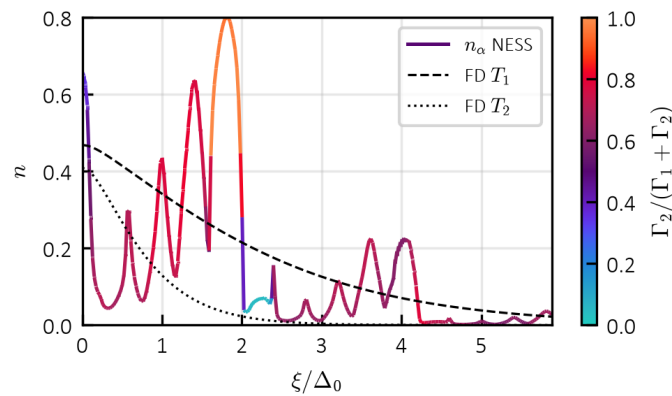
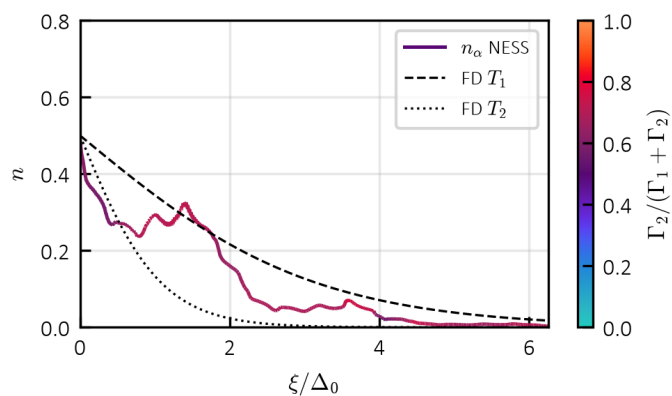


4. $A_{\xi}(\epsilon, \omega) \rightarrow \delta(\epsilon \pm \omega)$ limita:

dve kopeči: $\Delta_0 = 1.0$, $\omega_1 = 2.2$, $\omega_2 = 1.8$

točka: $T_z = 1.0$, $n_s = 367$.

(A) $T_1 = 2.905$, $\Delta = 0.0$, $m = -0.575$ (B) $T_2 = 2.900$, $\frac{\Delta}{\Delta_0} = 0.1989 \approx 0.2$, $m = -0.41$

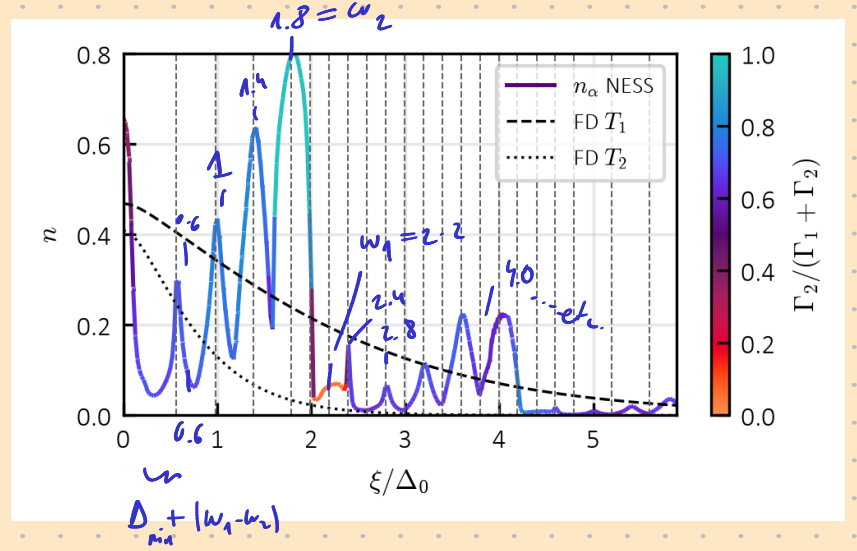


$$\mathcal{E} \rightarrow \mathcal{E} + a\omega_1 + b\omega_2, \quad a, b \in \mathbb{Z}$$

$$|\omega_1 - \omega_2| = 0.4$$

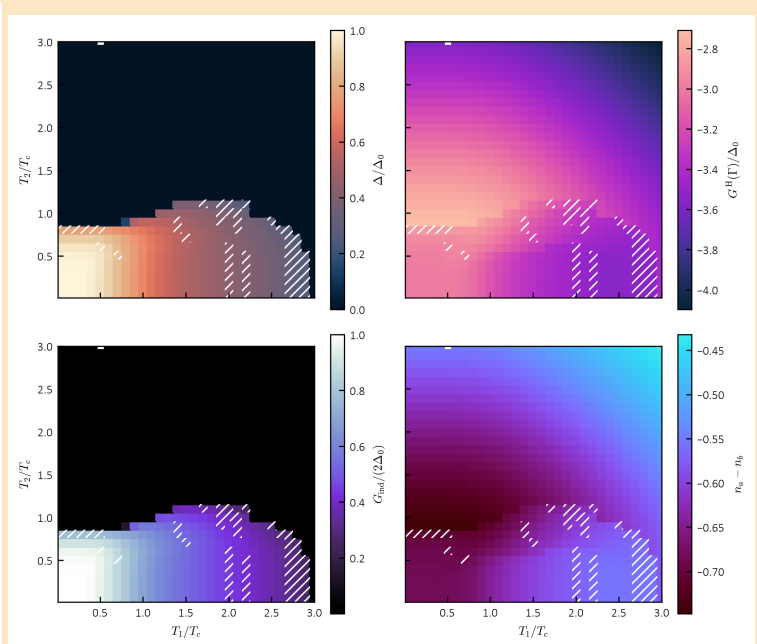
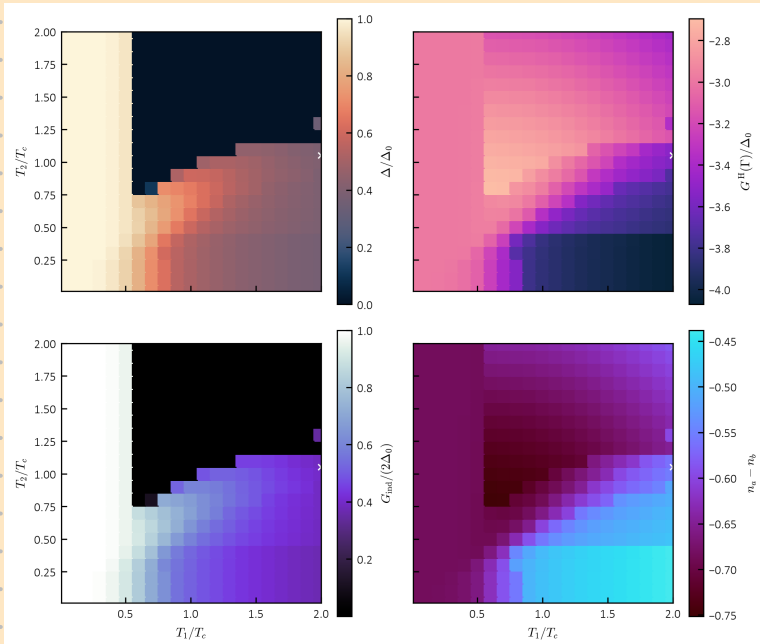
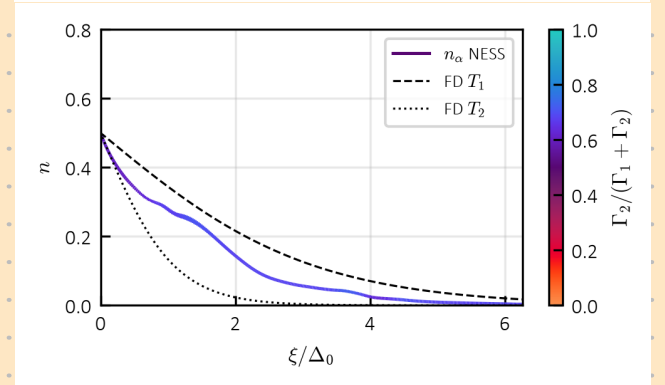
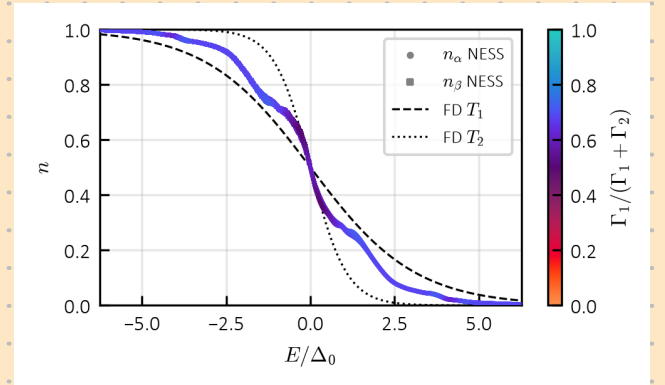
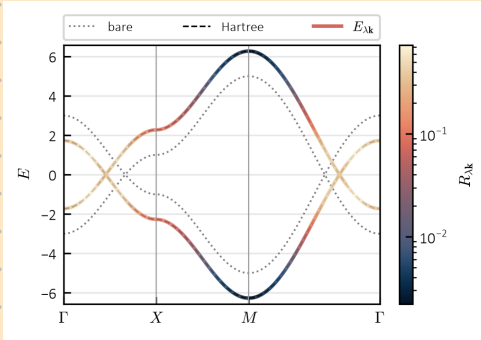
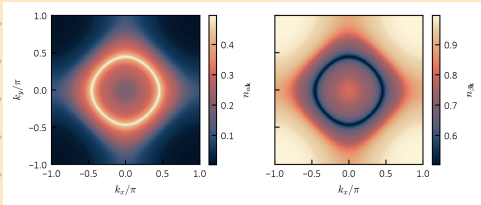
$$2\Delta_{\min} = \omega_1 - \omega_2 + \underbrace{\text{bandwidth}}_{\omega_s \approx 0}$$

$$\Delta_{\min} \approx 0.2 \quad \checkmark$$

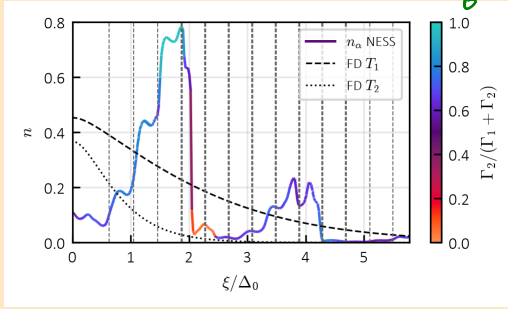


$$\gamma \in \mathcal{S} \rightarrow \mathcal{A} \quad \gamma = 0.1, \quad \mathcal{Z}_\gamma(\mathcal{E}, \nu)$$

$$\nu T_1 = 2.9: \text{ gap ityke.}$$



↖ hierin $\eta > 0$, fixer width kernel zu grid.



Kubo delung: Γ_{IJ} :

bin weight:

$$\sum_{E \in \mathcal{E}, \vec{p} \in \mathcal{P}} \delta(E_{\vec{p}} - E_{\vec{p}} - \omega_m) \rightarrow \widetilde{W_I} W_J \chi_{I,J}(E - \omega_m)$$

$$\Gamma_{I \leftarrow J} = 2\pi |g|^2 A_m(I, J) \left[(1 + N(\omega_m)) L_{IJ}(E - \omega_m) + N(\omega_m) L_{IJ}(E + \omega_m) \right]$$

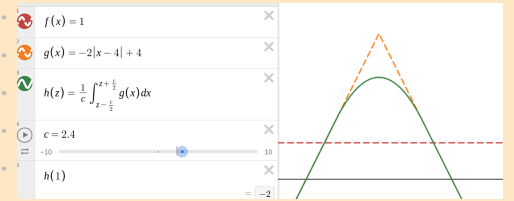
$$\Gamma_{J \leftarrow I} = e^{-\epsilon \hbar} \Gamma_{I \leftarrow J} \text{ (we } e^{-\omega_m \hbar}, \text{ da drehen DB).}$$

$$L_{IJ}(y) = \frac{1}{c} \int_{b - \frac{c}{2}}^{b + \frac{c}{2}} \tau_{h_I, h_J}(u) du$$

frühzeit, konvolution durch stärke:

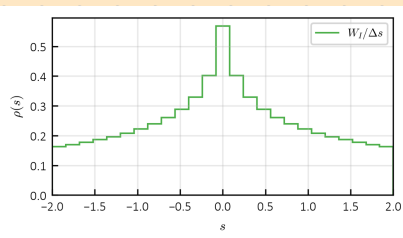
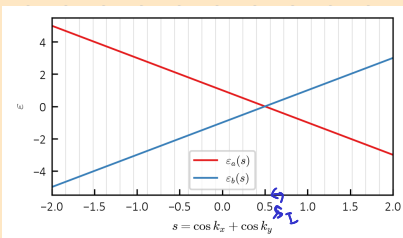
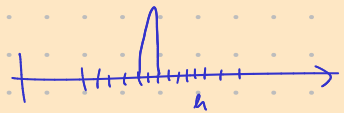


$$c = \max\left(\frac{h_I + h_J}{2}, c_{\min}\right) \quad c_{\min} = \max(h_{\max}, \text{Strich } \eta)$$



$\eta > 0$: $c = \eta$:

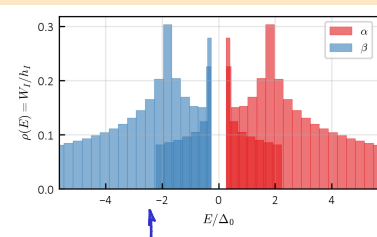
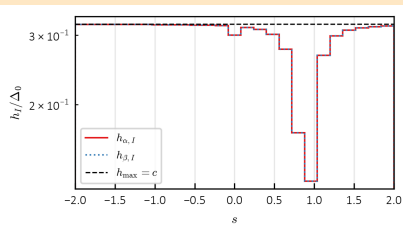
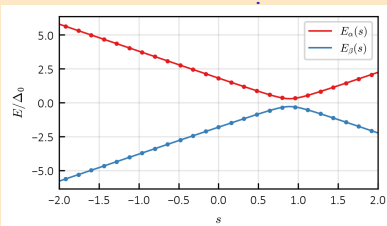
$$L_{IJ}(y) \rightarrow \frac{1}{\eta} \cdot \text{Box}\left[|y| < \frac{\eta}{2}\right]$$



$$p(s) = \int \frac{d^2 k}{(2\pi)^2} \delta(r - (c \cos k_x - c \cos k_y)) ds$$

$$W_I = \int \frac{d^2 k}{(2\pi)^2} \Theta((c \cos k_x + c \cos k_y) \in [s_I, s_{I+1}]) \frac{1}{s_I}$$

$$p(s_I) = \frac{W_I}{\Delta s_I}$$



$$\delta(f(x)) = \sum_{s_0: f(s_0)=0} \frac{\delta(x - s_0)}{|f'(s_0)|} \rightarrow p(E) = \frac{p(0)}{|f'(E)|}$$

